Double-Date Campaign Performance Analysis

Did stronger demand justify the campaign costs?

Synthetic E-Commerce Dataset Analysis
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Bryan Christian
Data Analyst & Business Planning Enthusiast

BigQuery SQL · Excel · Canva

Business Question ?

The decision: **expand**, **redesign**, or **limit Double-Date**?

Business Context

Double-Date campaigns use increased **exposure, discounts, vouchers, cashback, and shipping support** to attract **more visits and purchases**. The key question is whether the **added sales** still leave **healthy contribution** after **campaign costs.**

Campaign Analyzed

9.9 · 10.10 · 11.11 · 12.12

Decision Framework

01 : Expanded → **Sales** and **contribution** both improve

02 : Redesign → **Sales** improve, but **contribution** declines after campaign cost

03 : Limit → **Sales** remain weak and **contribution** declines

To answer the business question fairly, campaign performance must be compared with a relevant Normal Days baseline.

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Scope of Analysis

This case evaluates **short-term demand and contribution only**. Long-term customer retention is outside the current analysis window.

Data & Method

A fair comparison: campaign days vs the rest of the same month

Main Formula

Average Daily Metric = Total Performance Period / Number of Days

Notes: Raw totals would be misleading because period lengths are different.

Comparison periods

Campaign	Campaign period	Normal Days baseline
9.9	Sep 1-9	Sep 10-30
10.10	Oct 1-10	Oct 11-31
11.11	Nov 1-11	Nov 12-30
12.12	Dec 1-12	Dec 13-31

Core table data

Business Data	Warehouse table	Total Row
Website/app visits	fact_sessions	973K+
Customer journey steps	fact_funnel_events	2.228M+
Completed purchases	fact_orders	64K+
Product/SKU Sales	fact_order_items	102K+

Notes: For the complete datasets, see at appendix (page 12)

Notes

- All the data is created sythentic by **LLM in personal Google Colab.**
- Proses Breaking down the data into several format table & visualization, all used **BigQuery & Microsoft Excel.**
- All the definition about **main table & metrics** will be shown at appendix.

Did Double-Date increase daily orders and TPV?

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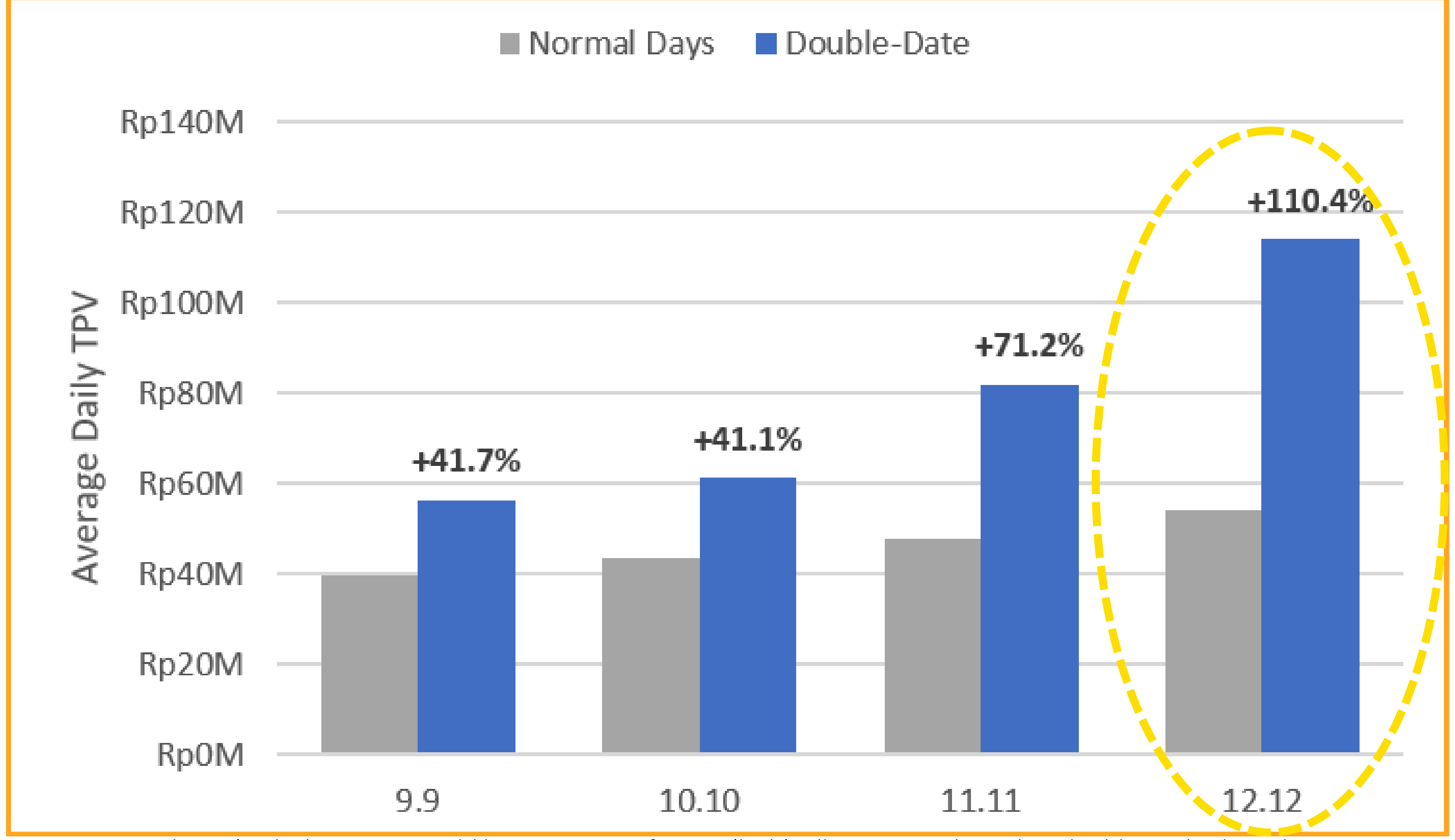
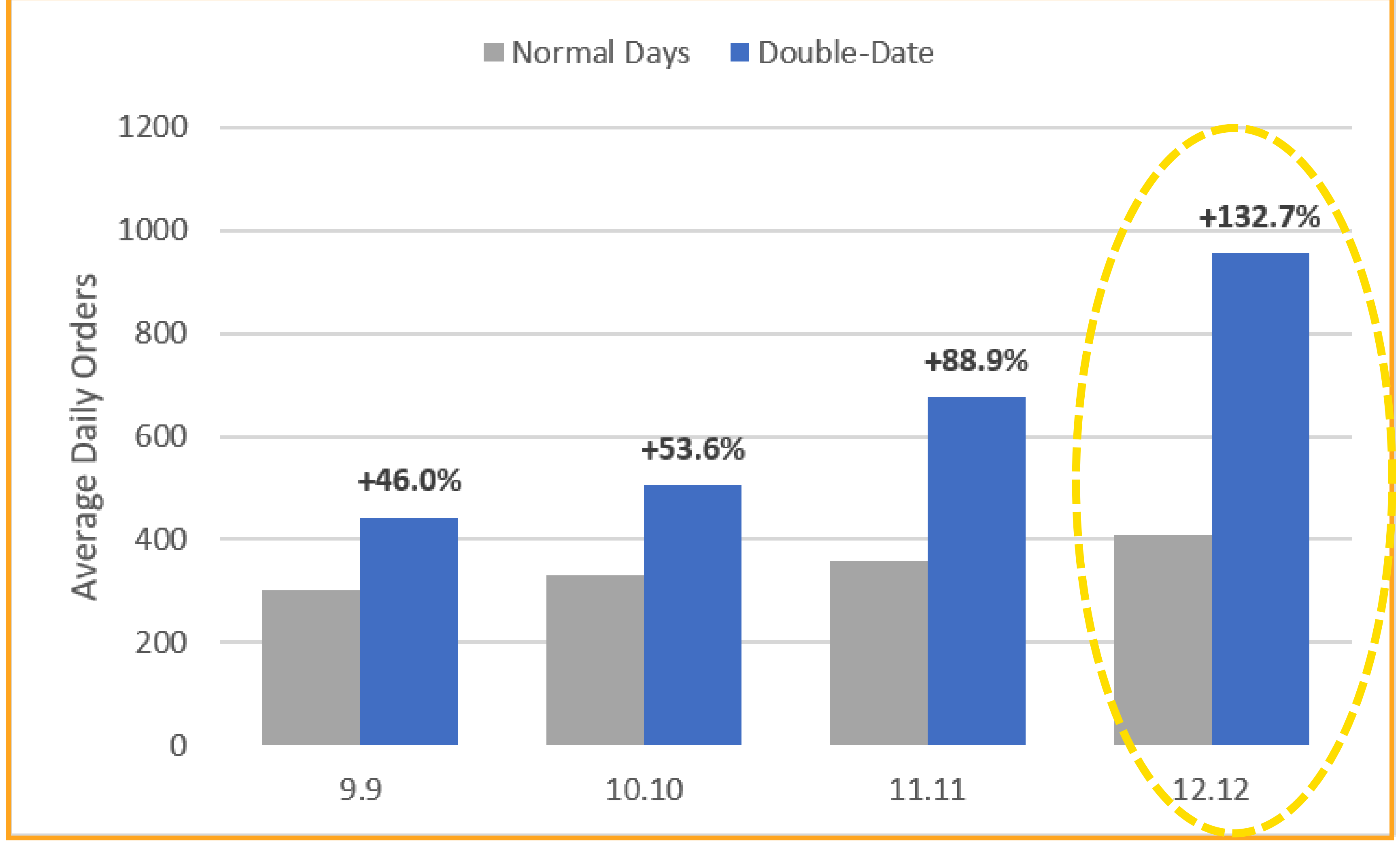
Chapter 1 · Demand Growth

Double-Date lifted daily orders and transaction value, led by 12.12

Average Daily Transaction Value



Average Daily Orders



Notes: Transaction value is the amount paid by customers after applicable discounts and vouchers in this synthetic project.

Core Highlights

12.12 AT A GLANCE

- **Daily Transaction Value:**
  - Rp54.2M → **Rp114.0M (+110.4%)**
- **Daily Order:**
  - 410 → **955 (+132.7%)**

What it Means?

Campaign impact became **stronger** toward **12.12.** Orders grew faster than transaction value.

What drove the growth: visits, conversion, or spending?

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Chapter 1 · Growth Drivers

More visits and higher purchase conversion drove growth; spending per order fell

Compact Performance Table

Campaign	Visits	Purchase conversion	Orders	Average order value	Transaction Value
9.9	+19.2%	+1.2 pp	+46.0%	-3.0%	+41.7%
10.10	+22.5%	+1.4 pp	+53.6%	-8.1%	+41.1%
11.11	+39.4%	+2.1 pp	+88.9%	-9.4%	+71.2%
12.12	**+56.2%**	**+2.9 pp**	**+132.7%**	**-9.6%**	**+110.4%**

Notes: Purchase conversion = percentage of visits ending in a completed purchase. | Average order value = transaction value divided by completed orders.

How 12.12 Was Build?

Visits +56.2% + Purchase Conversion +2.9 pp = Orders +132.7%

Orders +132.7% + AOV -9.6% = Transactions Value 110.4%

What it Means?

More people visited and completed purchases, but **average spending per order was lower.** Double-Date growth was **volume-led**, not driven by **larger baskets.**

Did the additional sales generate enough contribution after campaign costs?

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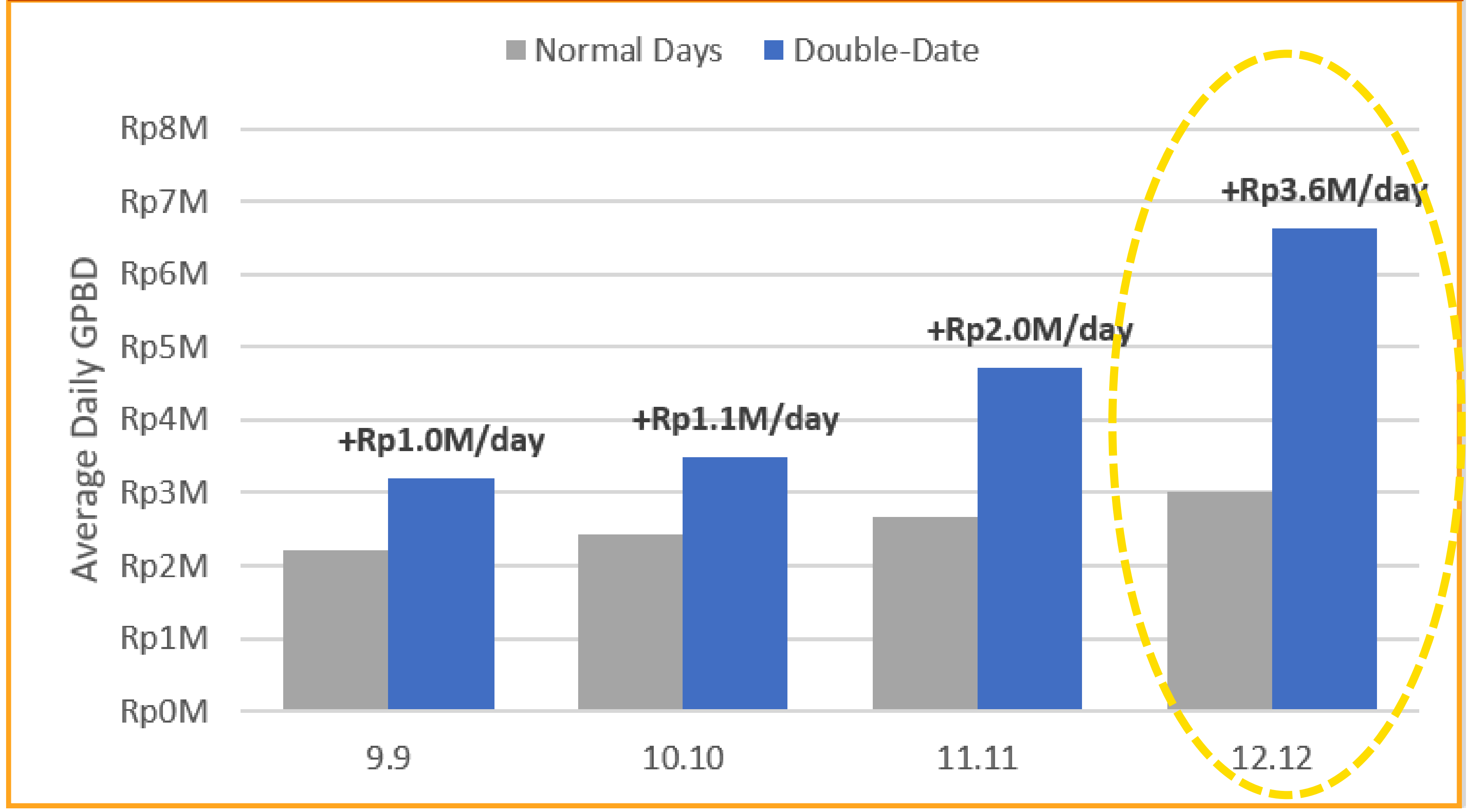
Chapter 2 · Financial Sustainability

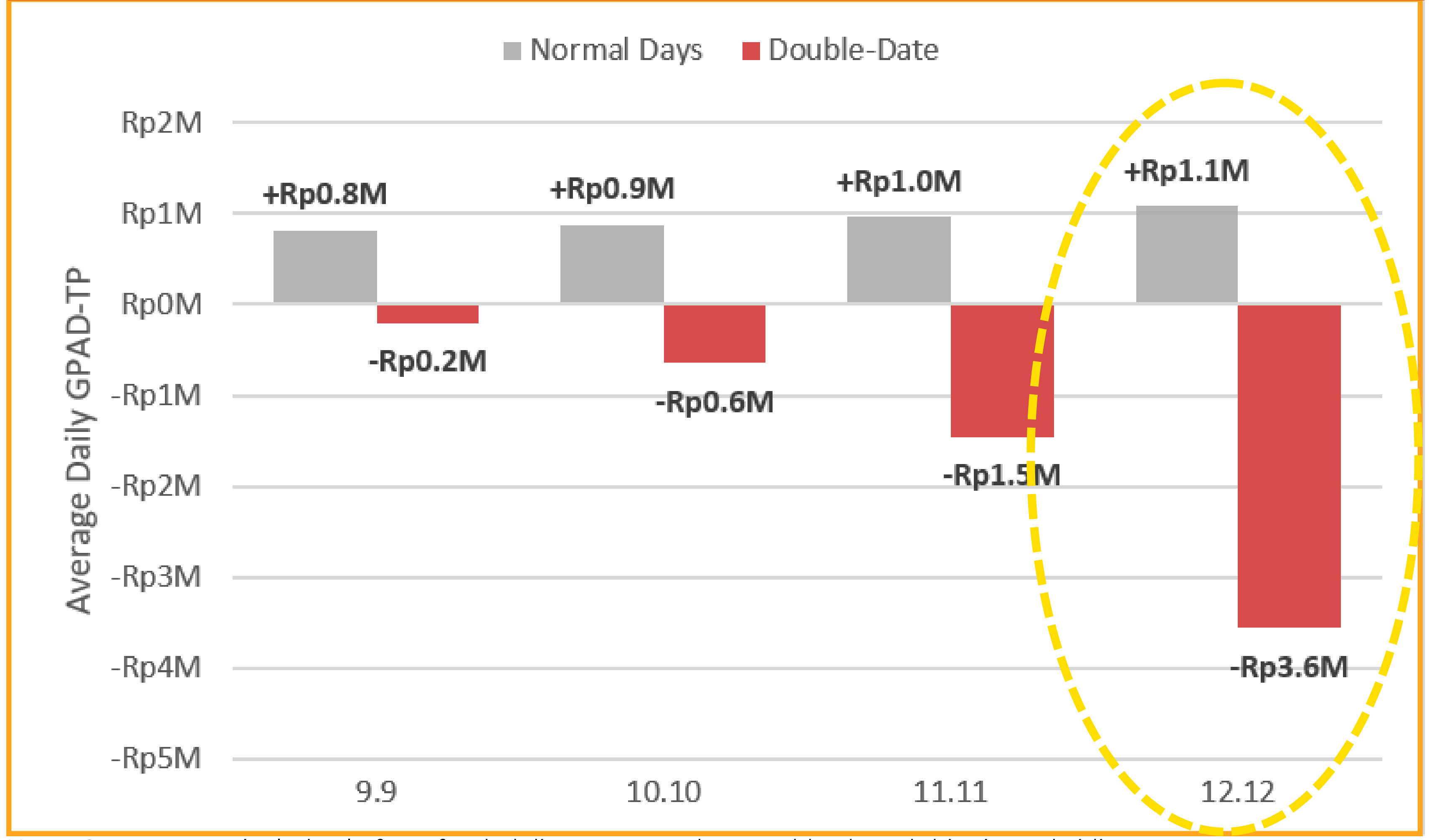
Campaigns generated more contribution before support costs, but after-cost contribution turned negative

Contribution before Support Costs



Notes: Contribution is the amount remaining from transaction revenue after selected costs. It is not the company's total net profit.

Contribution After Support Costs



Notes: Support costs include platform-funded discounts, vouchers, cashback, and shipping subsidies.

Core Highlights: 12.12 AT GLANCE

Before-cost contribution
Rp3.0M → **Rp6.6M/day**
Improvement: +Rp3.6M/day

After-cost contribution
Rp1.1M → **-3.6M/day**
Gap vs Normal Days: -4.64M/day

Which campaign created the largest after-cost shortfall?

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What Changed?

Platform-funded discounts, vouchers, cashback, shipping support **exceeded** the additional contribution generated by higher sales.

Chapter 2 · Priority Diagnosis

12.12 delivered the strongest sales growth—and the largest contribution shortfall

Compact Performance Table

Campaign	Before-cost margin: Normal → Campaign	After-cost margin: Normal → Campaign	Daily shortfall vs Normal Days	Estimated total campaign shortfall
9.9	5.57% → 5.68% +0.11 pp	2.05% → -0.35% -2.41 pp	Rp1.02M	Rp9.14M
10.10	5.58% → 5.68% +0.10 pp	2.02% → -1.06% -3.08 pp	Rp1.52M	Rp15.22M
11.11	5.57% → 5.74% +0.17 pp	2.02% → -1.78% -3.81 pp	Rp2.43M	Rp26.72M
12.12	**5.57% → 5.81% +0.24 pp**	**1.99% → -3.13% -5.12 pp**	**Rp4.64M**	**Rp55.71M**

Where did 12.12 perform well, and what created the shortfall?

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WHY FOCUS ON 12.12?

Strongest Demand Growth

- **Orders: +132.7%**
- **TPV: +110.4%**

Largest Margin Decline

- **−5.12 percentage points**

Largest Estimated Shortfall

- **Rp55.71M**

Main Decision

Select 12.12 for deeper funnel, category, and campaign-cost analysis.

Chapter 3 · Customer Journey

12.12 converted more visits into purchases, especially from cart to checkout

Funnel Journey

Funnel-step improvement chart 12.12

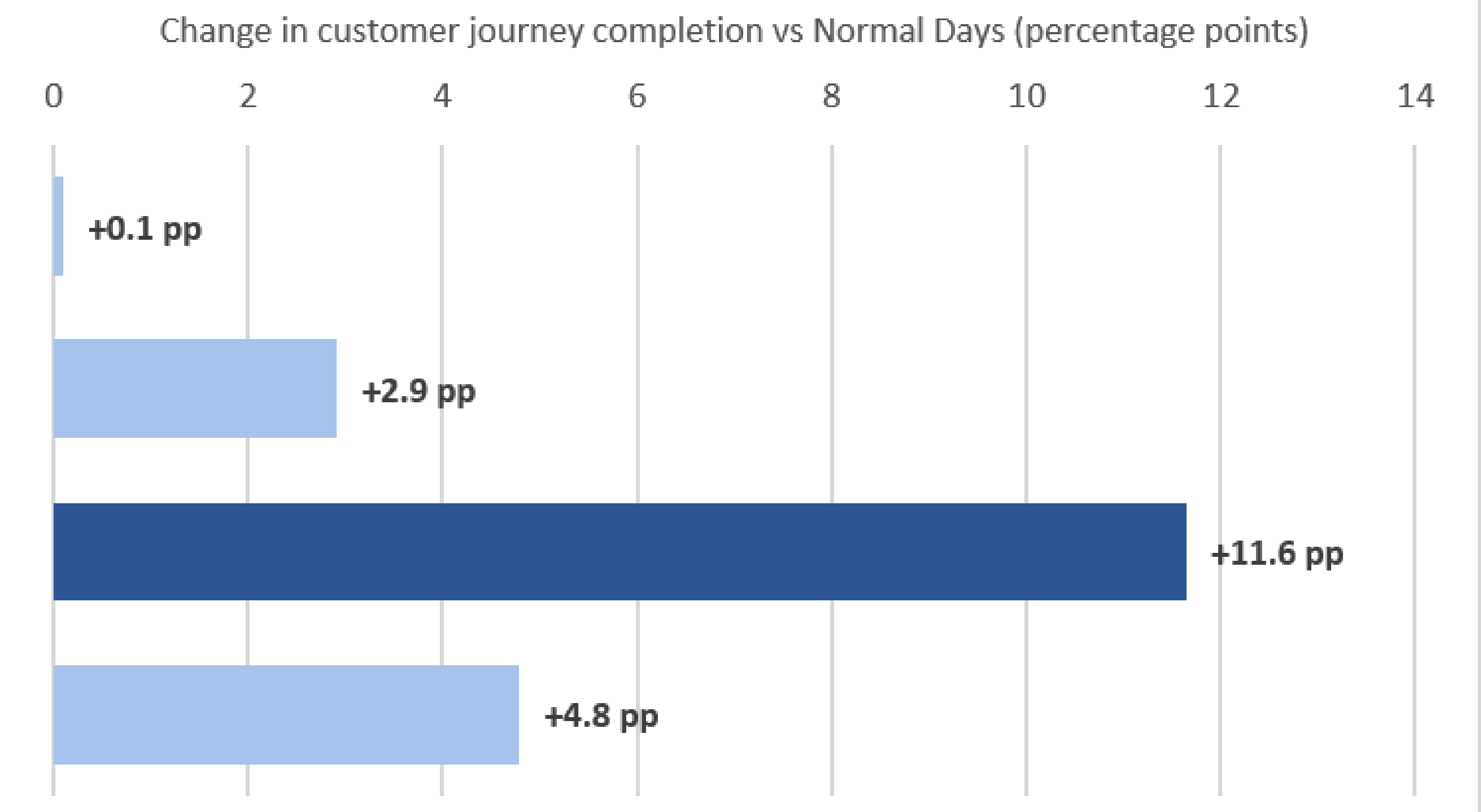
Stage

Visit → Product View

Product View → Add to Cart

Add to Cart → Checkout

Checkout → Purchase



Overall Result

- Purchase Conversion Vs Normal Days:
  - 5.94% → **8.89%**
  - **+2.95 percentage points**

MAIN IMPROVEMENT

- Cart → Checkout
  - **+11.6 percentage points**

Notes: Purchase conversion is the percentage of visits ending in a completed purchase.

What it Means?

Customers who had **already added products to their carts** were more likely to **continue to checkout.**

If conversion improved, was the financial shortfall concentrated in only a few product categories?

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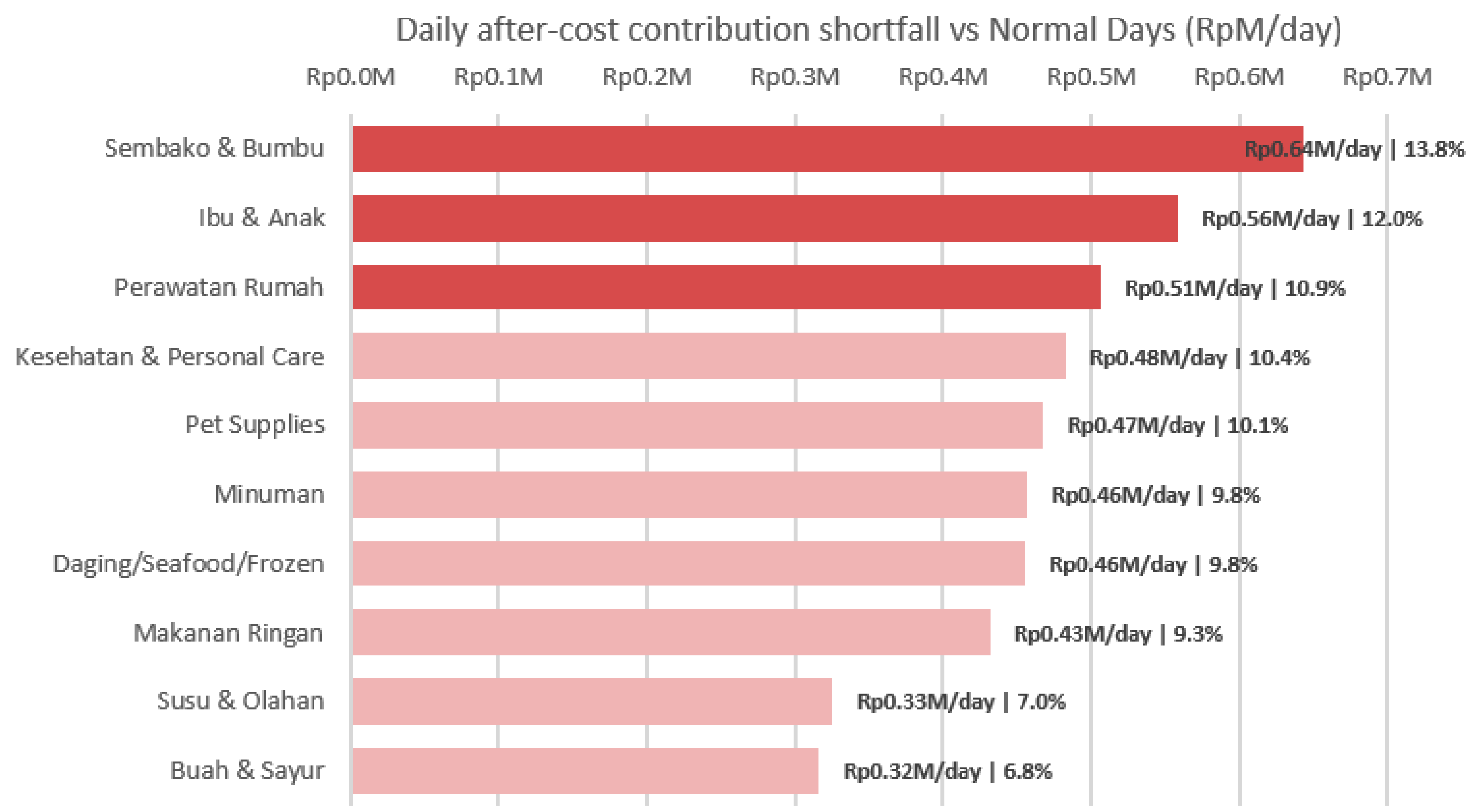
Chapter 3 · Category Diagnosis

The contribution shortfall was broad-based, not driven by one category

Horizontal category shortfall chart



Overall Result

BREADTH OF THE ISSUE

- **10 of 10 categories recorded lower after-cost contribution.**

LARGEST CATEGORY

- **Sembako**
  - **Rp0.64M/day shortfall**
  - **13.8% of total categories**

Key Interpretation

- **The top three categories explained only 36.8%** of the total shortfall. **No category dominated the result.**
- **No single category** explained most of the financial decline. **Focusing only on Sembako or one merchant** would provide an **incomplete diagnosis.**

If the shortfall affected every category, which common campaign costs created it?

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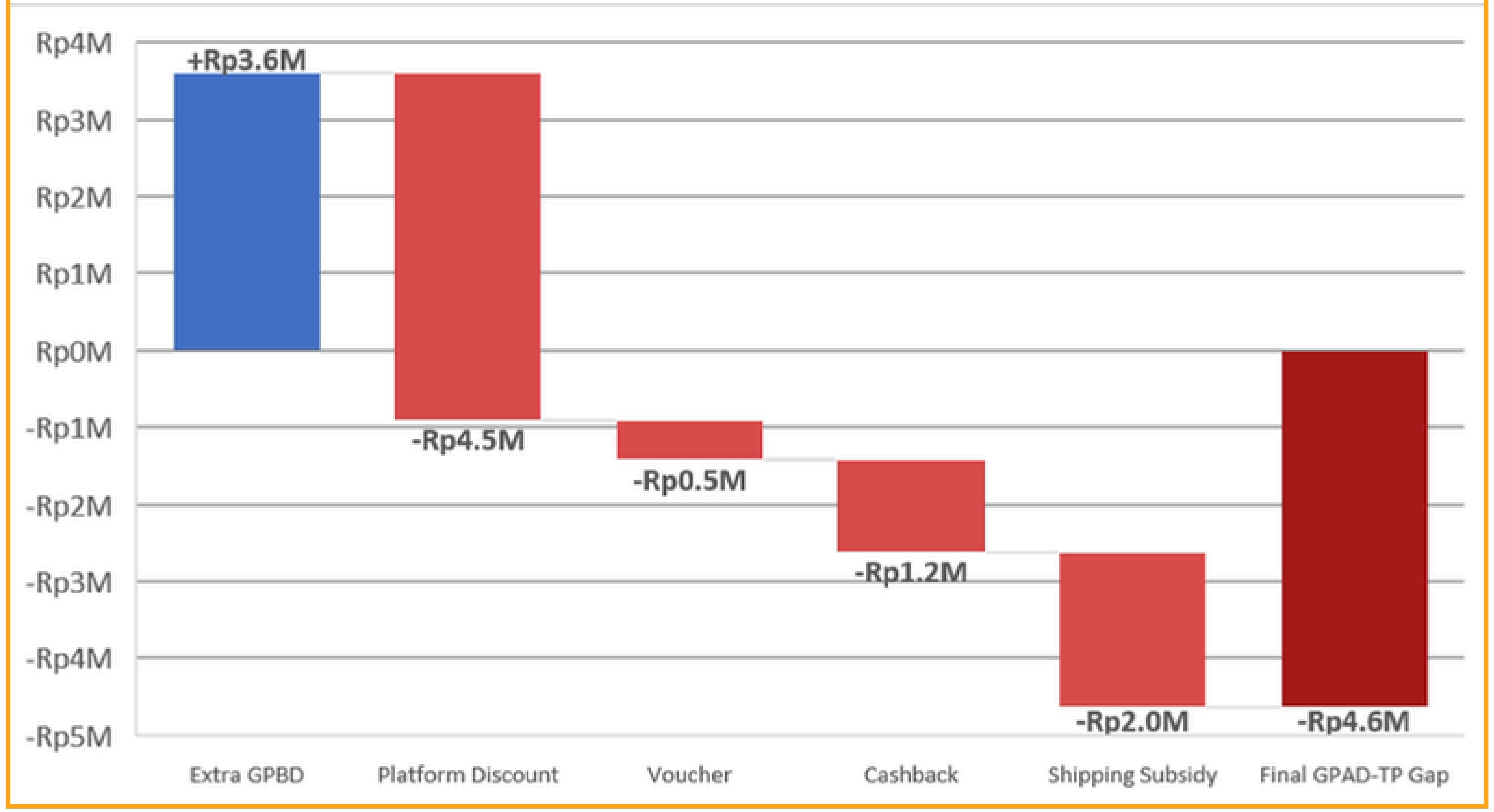
Chapter 3 · Campaign Cost Diagnosis

Campaign support costs were 2.29× the extra contribution generated

Cost Waterfall Chart



Notes: All values represent the difference between 12.12 and Normal Days, averaged per day.

The Calculation

Extra GPBD / Extra contribution before support costs
+Rp3.61M/day

Additional Support Costs (Platform Discount, etc)
-Rp8.25M/day

Final GPAD-TP / After Support Costs Shortfall
-Rp4.64M/day

Largest Cost Drivers

Platform discount: 54.6%

Shipping support: 24.6%

Combined: 79.2%

What it Means?

The campaign successfully **increased demand and purchase conversion**. However, its **platform-funded support costs exceeded** the additional contribution generated.

Redesign campaign support while preserving the demand growth.

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Final Recommendation

Redesign campaign support before selectively expanding Double-Date

Demand Performance

01 : Orders → + 132.7%

02 : Transaction value → +110.4%

03 : Conversion → +2.95 percentage point

Financial Performance

01 : Contribution after costs performance downfall → -Rp4.64M/day

02 : Platform Discount + Shipping Support → 79.2% of additional campaign support costs

03 : Support Costs / Extra Contribution → 2.29x

Final Decision

REDESIGN BEFORE EXPANDING

Three Action Areas

1. **Preserve demand growth:** Preserve the **campaign elements** associated with **higher visits and checkout completion**. Continue monitoring **purchase conversion** to ensure the redesigned campaign **does not weaken customer demand.**

2. **Reduce the largest cost drivers:** Target **platform discounts** instead of applying the same depth broadly. Review **minimum-spend and eligibility rules for shipping support**. **Test merchant co-funding** where commercially feasible.

3. **Pilot before expanding:** Run a **controlled pilot using category-level cost limits**. Compare the **redesigned campaign** with a **relevant baseline** before **expanding** it across more categories.

Proposed Scale-Up | **Positive demand uplift | After-cost margin >= 0% | Support Cost / Extra Contribution <= 1.0x**

Appendix A1 · End-to-End Project Workflow

Google Colab → Generate synthetic data

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CSV / Parquet Archive → Store reproducible source tables

▼

Google BigQuery → Load, join, and aggregate tables using SQL

▼

Microsoft Excel → Validate outputs and create charts

▼

Canva → Build the analytical story and presentation

Appendix A2 · Fact Table Dictionary

Table	Grain	Main Analytical Use
fact_sessions	one row per visit/session	Visits, traffic channel, device
fact_funnel_events	One row per customer journey event	View, cart, checkout, purchase
fact_orders	One row per completed merchant-level order	Orders, TPV, GPBD, GPAD-TP
fact_order_items	One row per order line/SKU	Product quantity, discounts, category
fact_inventory_daily	One row per date–SKU	Availability, viewable SKU, OOS
fact_returns	One row per returned order item	Return quantity, reason, refund

Appendix A3 · Demand and Growth Metric Definitions

Metric	Formula	Simple Meaning
Sessions	COUNT(DISTINCT session_id)	Number of visits
Orders	COUNT(DISTINCT order_id)	Number of completed purchases
Purchase conversion	Orders/Sessions	Share of visits ending in purchase
Average Order Value (AOV)	Total TPV/Orders	Average customer spending per order
Daily TPV	SUM(TPV) per day	Customer-paid transaction value per day
Uplift (Growth)	(Campaign − Normal) ÷ Normal	Relative change against baseline
Percentage-point change	Campaign rate − Normal rate	Absolute difference between two rates
Funnel-step rate	Next-step sessions ÷ Previous-step sessions	Share continuing to the next step

Appendix A4 · Financial Metric Definitions

Metric	Formula
GMV	List price × Quantity
Customer-paid TPV	GMV − Seller discount − Platform discount − Voucher
Commission revenue	(GMV − Seller discount) × Commission rate
GPBD	Commission revenue + Service fee revenue − Payment fee
GPAD-TP	GPBD − Platform discount − Voucher − Cashback − Shipping subsidy
GPBD margin	Total GPBD/Total TPV
GPAD-TP margin	Total GPAD-TP ÷ Total TPV
Daily contribution shortfall	Normal Days daily GPAD-TP − Campaign daily GPAD-TP
Support-cost ratio	Additional campaign support cost ÷ Extra GPBD generated

Appendix A5 · Project Resources

[Click Here for Complete Analysis]